Jin Heo

jinheo44@gmail.com • https://www.linkedin.com/in/jinheo4 • https://jheo4.github.io

Summary

- Senior Researcher at Dolby and PhD in Computer Science from Georgia Tech, specializing in ML inference serving and adaptive resource management for real-time systems. Research and engineering experience spans resource-constrained edge and scalable cloud environments, from algorithm design for XR and cloud gaming to production-scale ML and live streaming infrastructure.

Experience

Dolby Laboratories — Atlanta, GA
Senior Researcher — Aug 2025 - Present

- Researching an effectiveness-driven inference serving system for real-time streaming perception on resource-constrained servers, replacing static deadline-based scheduling with a mechanism that estimates the temporal effectiveness of inference results; reduces GPU utilization by 60–70% on average while maintaining comparable streaming average precision (sAP).
- Collaborating with Dolby ML researchers to integrate and deploy their models into a cloud cluster, architecting the end-to-end service workflow from ML inference through downstream processing and streaming.
- Designed a communication architecture for distributed multimedia microservices using KV caches and object storage for high-throughput data exchange and Pub/Sub for lightweight control-flow messaging; deployed via Kubernetes and Helm.
- Architected a scalable AWS-based cloud platform for sports live streaming, provisioning infrastructure with Terraform and integrating heterogeneous services across ingestion, processing, and delivery.
- Engineered resilient live-streaming telemetry pipelines (CMCD/CMSD) for Dolby OptiView's NFL streaming service, sustaining ~1M requests within 15 minutes with zero data loss through optimized ingestion windows and automated failure recovery workflows.

Georgia Institute of Technology — Atlanta, GA
Graduate Research Assistant — Aug 2019 - Aug 2025

- Developed Stimpack, an adaptive resource management system for scalable multi-user cloud gaming that balances server-side rendering cost against user-perceived quality, achieving up to 24% higher service quality and 2x user capacity with the same resources; published at NSDI 2026 [1].
- Designed a low-latency, soft real-time inference serving system for XR perception models, enabling scalable multi-user serving on resource-constrained edge servers [2].
- Built FleXR, a distributed stream processing system for flexible distribution of real-time XR workloads across edge-cloud environments, achieving up to 50% lower end-to-end latency and 3.9x higher pipeline throughput compared to alternatives [4, 7].

Dolby Laboratories — San Francisco, CA
Research Intern — May 2024 - Aug 2024

- Built a Universal Scene Description (USD) parser and partial-query retrieval method, enabling low-latency streaming of interactable 3D graphics for real-time content handling.

AT&T Labs — Austin, TX
Research Intern — May 2023 - Aug 2023

- Developed GT-Craft, a fast-prototyping framework for city-scale geospatial digital twins in Unity 3D, enabling simulation and analysis of AT&T base station wireless network performance [3].

Ericsson Research — Santa Clara, CA
Research Intern (remote, part-time) — Mar 2021 - Dec 2022

- Developed FLiCR, a lightweight LiDAR point cloud compression method for edge-assisted real-time perception, achieving a 12x compression ratio with up to 80% lower end-to-end latency than prior methods, along with an interpolation technique to recover compression-lost geometry that led to an international patent application (WO2024073084A1) [5, 6].

University of California, Irvine — Irvine, CA
Research Assistant (Undergraduate Research Opportunities Program) — Jul 2017 - Jan 2018

- Designed an FPGA acceleration framework for computer vision algorithms using OpenVX graph pipelines [8].

CSIRO (Commonwealth Scientific and Industrial Research Organization) — Brisbane, Australia
Research Assistant (Undergraduate Research Opportunities Program) — Sep 2016 - Feb 2017

- Developed and field-validated a network flooding algorithm for efficient broadcasting in battery-constrained Wireless Sensor Networks (WSNs).

Education

Georgia Institute of Technology Atlanta, GA

Doctor of Philosophy in Computer Science; GPA: 4.0 2019 - 2025

- Thesis: "Adaptively Serving XR Workloads from Resource-constrained Edge"

Ajou University Suwon, South Korea

Bachelor of Science in Computer Engineering; Summa Cum Laude 2016 - 2018

Publications

1. **Heo, J.**, Wang, V., Bhardwaj, K., & Gavrilovska, A. (2026). *Stimpack: An Adaptive Rendering Optimization System for Scalable Cloud Gaming*. In 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26) (pp. 397-414).
2. **Heo, J.** and Gavrilovska, A., 2024, December. Poster: *Adapting XR Perception Serving for Edge Server Scalability*. In 2024 IEEE/ACM Symposium on Edge Computing (SEC) (pp. 518-520). IEEE.
3. **Heo, J.**, Novlan, T., Akoum, S. and Gavrilovska, A., 2024, December. *GT-Craft: A Framework for Fast Prototyping Geospatial-Based Digital Twins in Unity 3D*. In 2024 IEEE/ACM Symposium on Edge Computing (SEC) (pp. 395-401). IEEE.
4. **Heo, J.**, Bhardwaj, K. and Gavrilovska, A., 2023, June. *FleXR: A system enabling flexibly distributed extended reality*. In Proceedings of the 14th Conference on ACM Multimedia Systems (pp. 1-13).
5. **Heo, J.**, Phillips, C. and Gavrilovska, A., 2022, December. *FLiCR: A fast and lightweight lidar point cloud compression based on lossy RI*. In 2022 IEEE/ACM 7th Symposium on Edge Computing (SEC) (pp. 54-67). IEEE.
6. **Heo, J.**, Phillips, G., Brodin, P.E. and Gavrilovska, A., 2022, December. Poster: *Making Edge-assisted LiDAR Perceptions Robust to Lossy Point Cloud Compression*. In 2022 IEEE/ACM 7th Symposium on Edge Computing (SEC) (pp. 293-295). IEEE.
7. **Heo, J.**, Bhardwaj, K. and Gavrilovska, A., 2021, December. Poster: *Enabling flexible edge-assisted XR*. In 2021 IEEE/ACM Symposium on Edge Computing (SEC) (pp. 465-467). IEEE.
8. Taheri, S., **Heo, J.**, Behnam, P., Chen, J., Veidenbaum, A. and Nicolau, A., 2018, April. *Acceleration framework for FPGA implementation of OpenVX graph pipelines*. In 2018 IEEE 26th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM) (pp. 227-227). IEEE.

Skills

- Programming & Systems: C, C++, Python, Linux, Git, CMake, Make
- Distributed Systems & Infrastructure: AWS, Kubernetes, Terraform, Helm, Docker, Flyte, Metaflow, gRPC, Kafka, Spark, Databricks, ZMQ
- ML & Perception: PyTorch, Scikit-learn, OpenCV, Point Cloud Library (PCL)
- Multimedia & Graphics: Universal Scene Description (USD), Unity, Unreal Engine, OpenGL, FFmpeg, GStreamer

Activities

- Academic Service: Program Committee (PC), ACM SIGOPS Annual Technical Conference (ATC) 2026, External Reviewer, IEEE OJ-COMS 2024.
- Teaching: Head Teaching Assistant, Advanced Operating Systems (CS6210/4210) at Georgia Tech (Spring 2022, Spring 2024).
- Open-Source Contributions: RaftLib (resolved pipeline scheduler bottlenecks for streaming efficiency); uvgRTP (implemented Linux build system support via pkg-config).